Document Layout Analysis with Graph-based Methods

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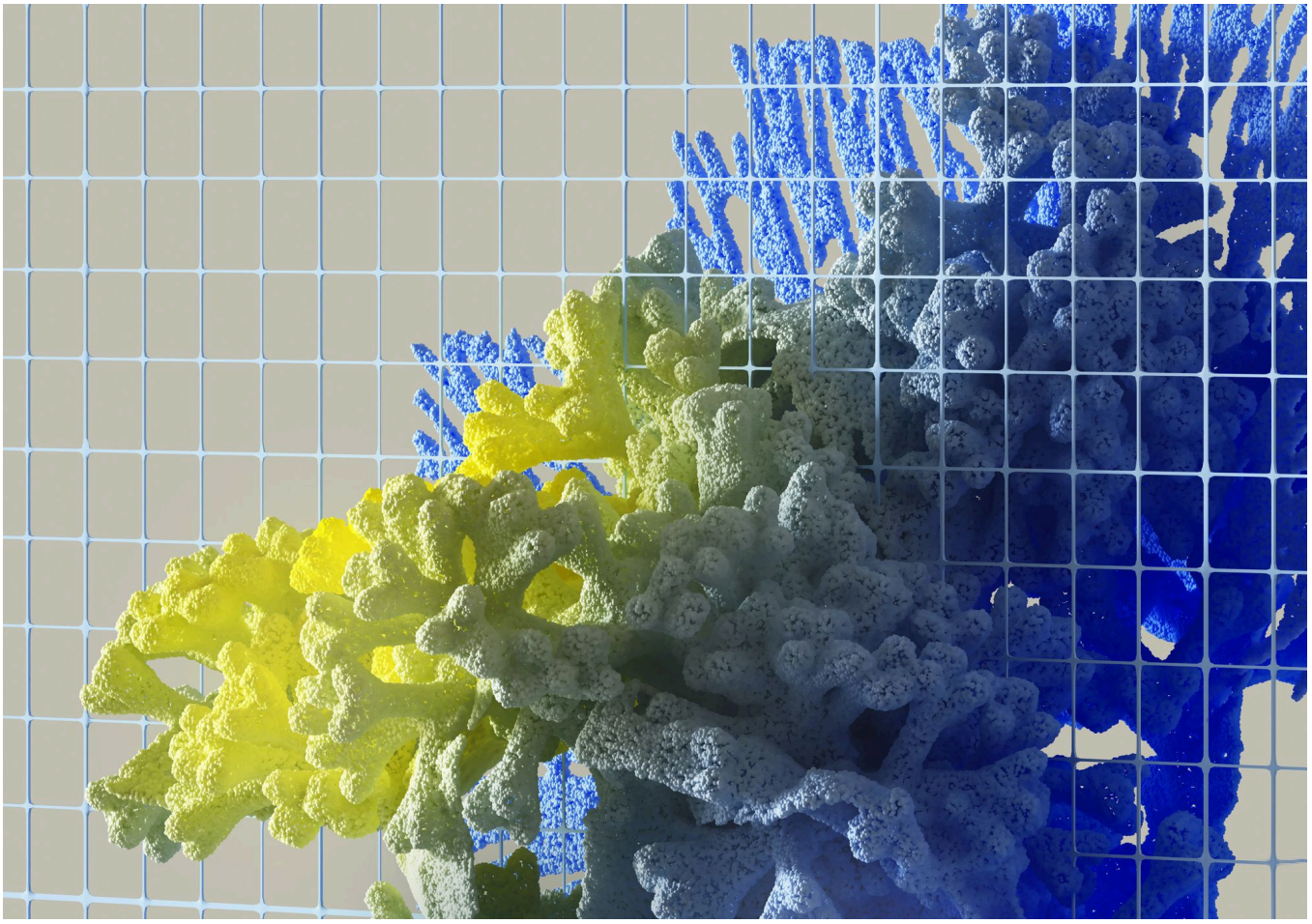


Enabling Document Intelligence through Lightweight, Structure-Aware Graph Models

06 / 05 / 2025

Introduction

Document intelligence (DI) is the application of artificial intelligence (AI) and machine learning (ML) technologies to analyze and extract information from documents. Research in document intelligence uses techniques from both natural language processing (NLP) and computer vision. Especially in industries like finance, legal services, etc., where a large number of documents are used and need to be analyzed to gain insights and make decisions based on them, document intelligence plays a crucial role in handling large amounts of data.

Document layout analysis (DLA) is a research area within the broader field of document intelligence. DLA is about understanding the structural and spatial arrangements of various components within a document. These components can include text blocks, headers, footers, images, tables, and more. DLA is used to identify the ordering and relation among these

components, such as deciding whether a piece of text under an image is a caption or a text block. Also, DLA can be a task in itself in some cases, such as object detection and text segmentation, or it can work as a precursor task to enable other document intelligence applications, such as document classification or information extraction. DLA is especially crucial for unstructured documents where there is little to no prior knowledge about the components and their placements in a document.

Document layout analysis can be used to great benefit in the financial sector, as it relies on a large number of documents in both back-office and front-office operations. Parsing and extracting information rapidly from these reports can be crucial in enabling stakeholders to derive actionable insights, comply with regulatory standards, and support decision-making based on comprehensive, well-structured information. To that end, we focused on two use cases: Environmental, Social, and Governance (ESG) reports released annually by companies and financial reports prepared in accordance with International Financial Reporting Standards (IFRS). ESG reports are prepared by each company in their own format, so they are unstructured even though the content of each report roughly contains similar information. Therefore, creating an information extraction system, even with the assistance of large-language models (LLMs), is a non-trivial task. That is why having a model that can automatically identify components and then classify them based on the information contained within can be crucial. The challenge with IFRS reports is different than ESG as they adhere to a common standard in layout and are more structured documents. However, these documents may contain large, sometimes nested, tables where it can be challenging to extract information from.

Figure 1: Example ESG report page in Turkish – It has a mixture of text boxes and image elements, and the format is unstructured.

SODAŞ SODYUM SANAYİİ A.Ş.

30 HAZİRAN 2021 VE 31 ARALIK 2020 TARİHLERİ İTİBARIYLA
ÖZET BİLANÇOLAR
(Tutarlar aksi belirtilmedikçe Türk Lirası ("TL") olarak ifade edilmiştir.)

			Bağımsız Denetimden Geçmiş
		Cari **Dönem**	Geçmiş Dönem
	Dipnot **Referansları**	30 Haziran 2021	31 Aralık 2020
VARLIKLAR			
Dönen Varlıklar			
Nakit ve Nakit Benzerleri	3	22.385.727	29.920.882
Ticari Alacaklar		3.292.548	1.735.674
- İlişkili Olmayan Taraflardan Ticari Alacaklar	5	3.292.548	1.735.674
Stoklar	7	4.928.954	4.959.805
Peşin Ödenmiş Giderler		525.965	452.764
- İlişkili Taraflara Peşin Ödenmiş Giderler	4	339.913	334.219
- İlişkili Olmayan Taraflara Peşin Ödenmiş Giderler	8	186.052	118.545
Diğer Dönen Varlıklar		92.274	9.346
TOPLAM DÖNEN VARLIKLAR		31.225.468	37.078.471
Duran Varlıklar			
Finansal Yatırımlar		-	94
Diğer Alacaklar		706.365	547.150
- İlişkili Olmayan Taraflardan Diğer Alacaklar	6	706.365	547.150
Maddi Duran Varlıklar	9	10.931.212	10.055.543
Kullanım Hakkı Varlıkları	9	155.759	122.421
Maddi Olmayan Duran Varlıklar		593.379	637.079
- Diğer Haklar		593.379	637.079
Ertelenmiş Vergi Varlığı		1.946.011	2.052.563
TOPLAM DURAN VARLIKLAR		14.332.726	13.414.850
TOPLAM VARLIKLAR		45.558.194	50.493.321

1 Ocak - 30 Haziran 2021 ara hesap dönemine ait özet finansal tablolar yayımlanmak üzere Sodaş Sodyum Sanayii A.Ş. Yönetim Kurulu tarafından 29 Temmuz 2021 tarihinde onaylanmıştır.

İzleyen dipnotlar, özet ara dönem finansal tabloların ayrılmaz parçasını oluşturur.

(1)

Figure 2: Example IFRS report page in Turkish – It has a more structured format, but it has nested tables

Document Layout Analysis Methods

Document layout analysis can be categorized into three main methods: image-based, text-based, and graph-based.

Image-based layout analysis focuses on processing the visual elements of a document, treating it as an image. **Text-based layout analysis** focuses on analyzing the textual content of a document, usually extracted using Optical Character Recognition (OCR). **Graph-based layout analysis** is based on representing the document as a graph, so that layout elements such as text blocks and images are represented as nodes, and various spatial or logical relations between these elements are represented as edges. Graph neural networks (GNNs) are utilized to predict the specific nodes and sub-graphs as layout elements and also to predict the relations between those elements through edges in graphs. Graph-based methods also use OCR and additional metadata information from documents to successfully build graphs.

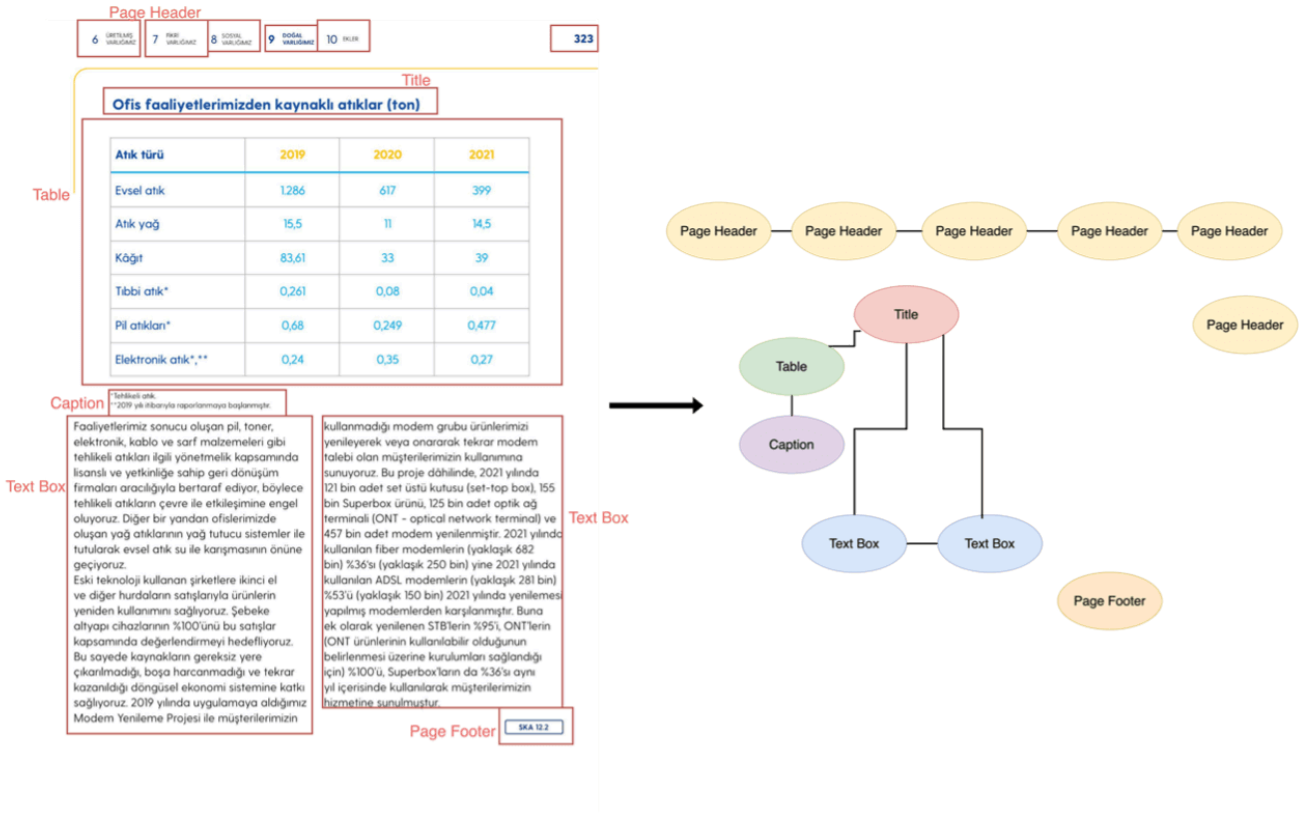


Figure 3: How a document page can be turned into a graph. Edges are shown only for highly correlated layout elements.

Graph-based methods generally require fewer computational resources than other methods because not every part of a page has to be encoded in a CNN or a transformer model. GNNs can encode graphs on a page with many fewer parameters, which also increases inference times. Additionally, since graphs are constructed from relationships between layout elements in documents, they can be used to identify relationships between components.

Graph Neural Networks

Graph neural networks (GNNs) are machine learning models designed to process data represented as graphs. Various data types can be represented as graphs, such as social networks (people as nodes and their connections as edges) or molecular structures (atoms as nodes and their bonds as edges). In the context of document layout analysis, each page of a document can be represented as a graph where nodes correspond to various layout elements (e.g., text blocks, tables), and edges capture the relationship among these elements, such as their distance or relative positions. GNNs can be used for performing various tasks: **graph-level tasks** (e.g., classifying an entire document), **node-level tasks** (classifying individual elements) and **edge-level tasks** (e.g., predicting hierarchical relations between nodes such as between an image and its caption).

For document layout analysis, node-level tasks are used to label each document element and edge-level tasks are used to connect same type of nodes as a single element and also to discover relations between different elements.

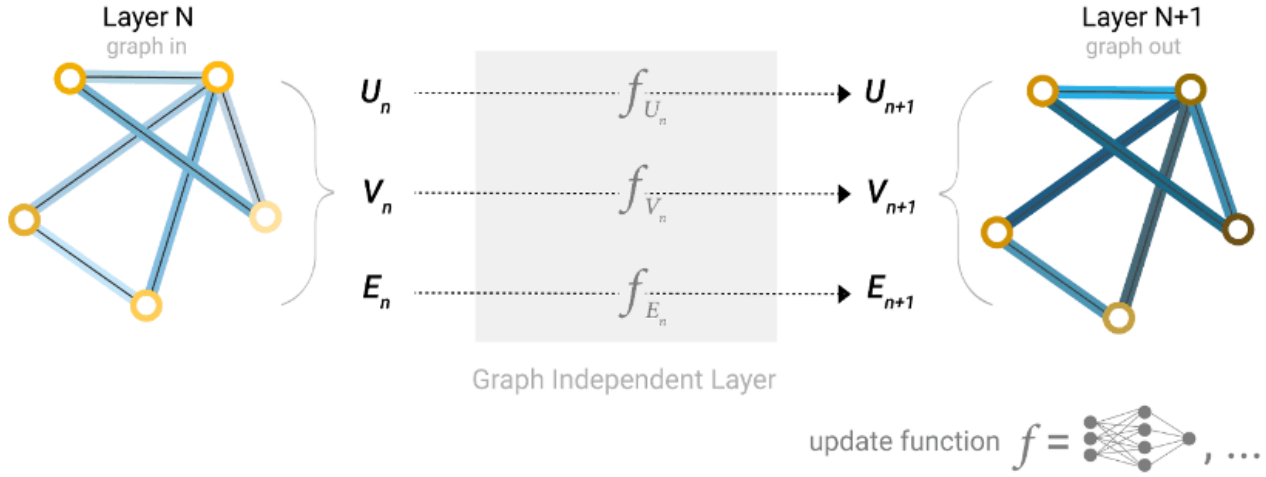


Figure 3A: A single GNN layer. Node, edge, and global weights are calculated and updated separately.

Document Layout Analysis as a Graph-Based Task

Layout elements of a document can be thought of as nodes of a graph, while spatial and logical relations between these elements can be considered as edges to construct a graph from a document page. Therefore, GNNs become a natural candidate for layout detection, where both types of layout documents and their relation to each other need to be analyzed. As a result, a significant number of recent studies have focused on GNN-based methods in DLA, often achieving performance close to or on par with state-of-the-art methods. At the same time, most of these studies use a similar architecture. So rather than listing all, three such approaches are chosen for this discussion to demonstrate different approaches in how to organize graphs, how to set node and edge properties and what sort of GNN architectures are utilized.

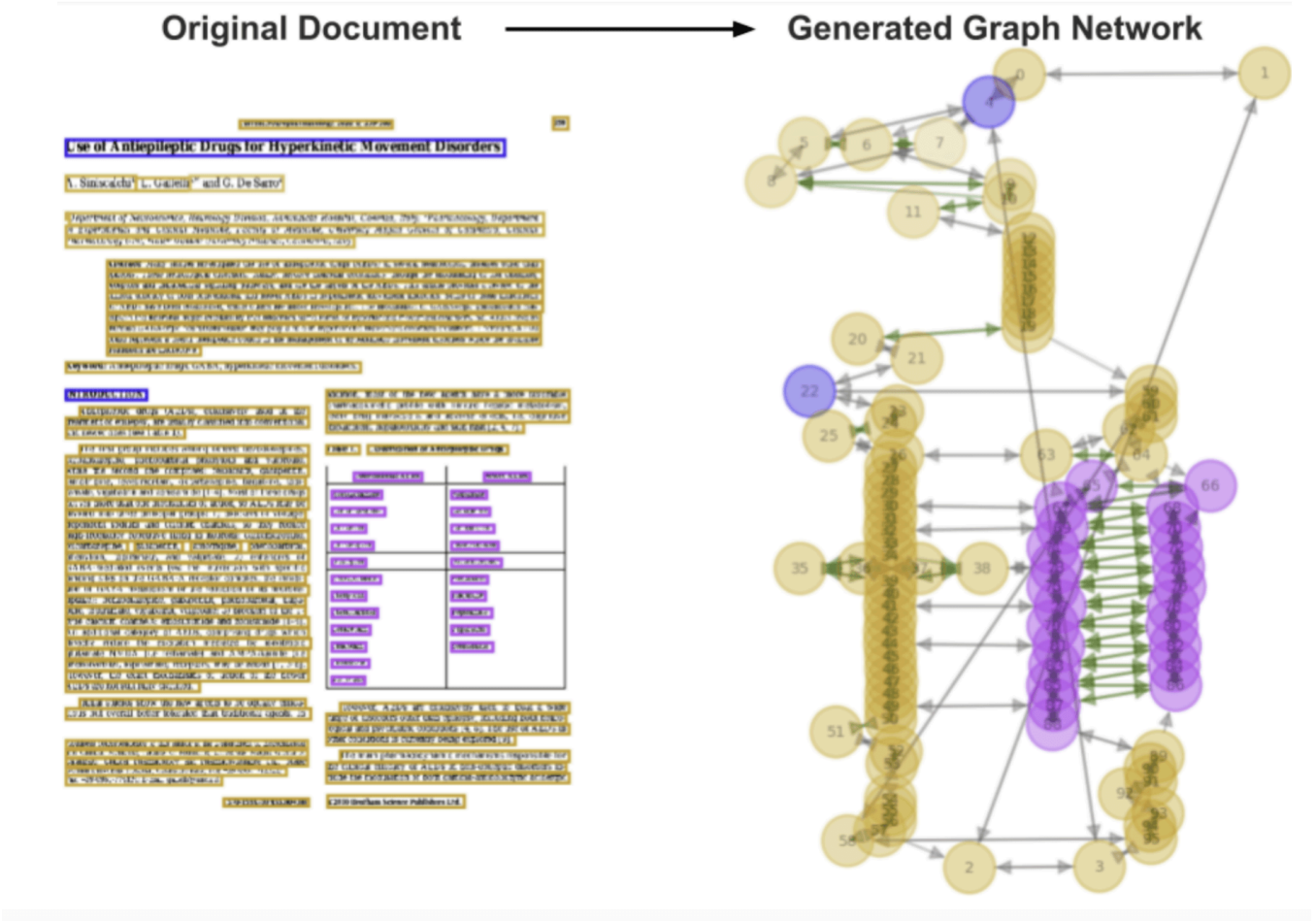


Figure 4: Constructing a graph from a page of a document

1. Graph-based Layout Analysis Model (GLAM)

GLAM is a lightweight Graph Neural Network (GNN) model for layout analysis. It has only 4 million parameters, while state-of-the-art image-based or text-based models have more than 100 million parameters. What makes this model lightweight is that it keeps its feature space limited by creating nodes and edges of a graph with mostly metadata information from PDF files. However, this brings the limitation that this model only works for PDF documents, as it does not use any visual features and instead relies on information provided from metadata.

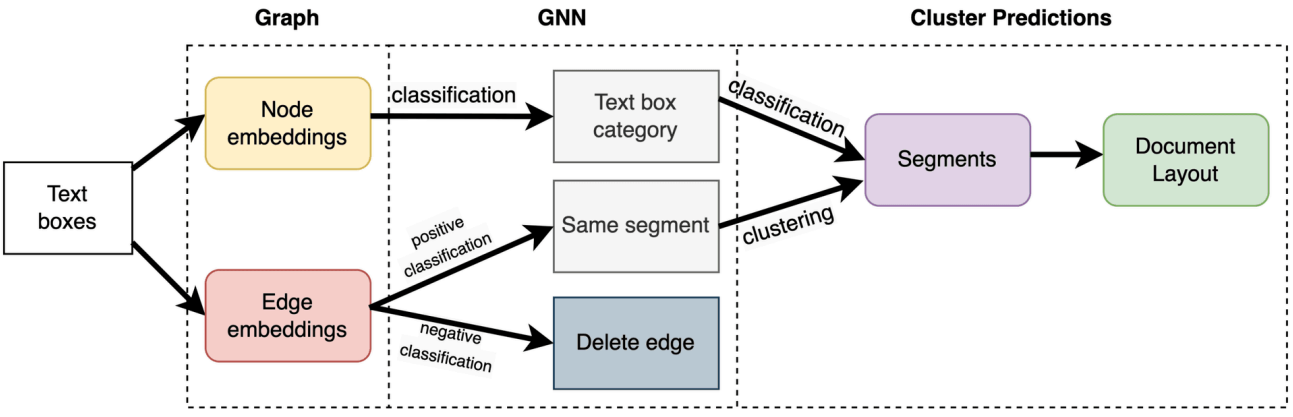


Figure 5: GLAM model pipeline

2. Paragraph2Graph

Paragraph2Graph is one of the better-performing graph-based layout models. Unlike GLAM, this model is designed to work with all documents. Thus, in order to create graph features, visual elements are extracted with a CNN, and the coordinates of text boxes are determined with OCR. Then the graph is built via visual and spatial features, and also with a Graph Sampling method, some edges are removed to make the graph sparse and easier to train. Specifically, the Dynamic Graph CNN (DGCNN) architecture is used as the GNN method for this architecture, as it allows the training of both node and edge embeddings. As a result, nodes can be classified to layout elements with a high accuracy and their relations to each other can also be captured.

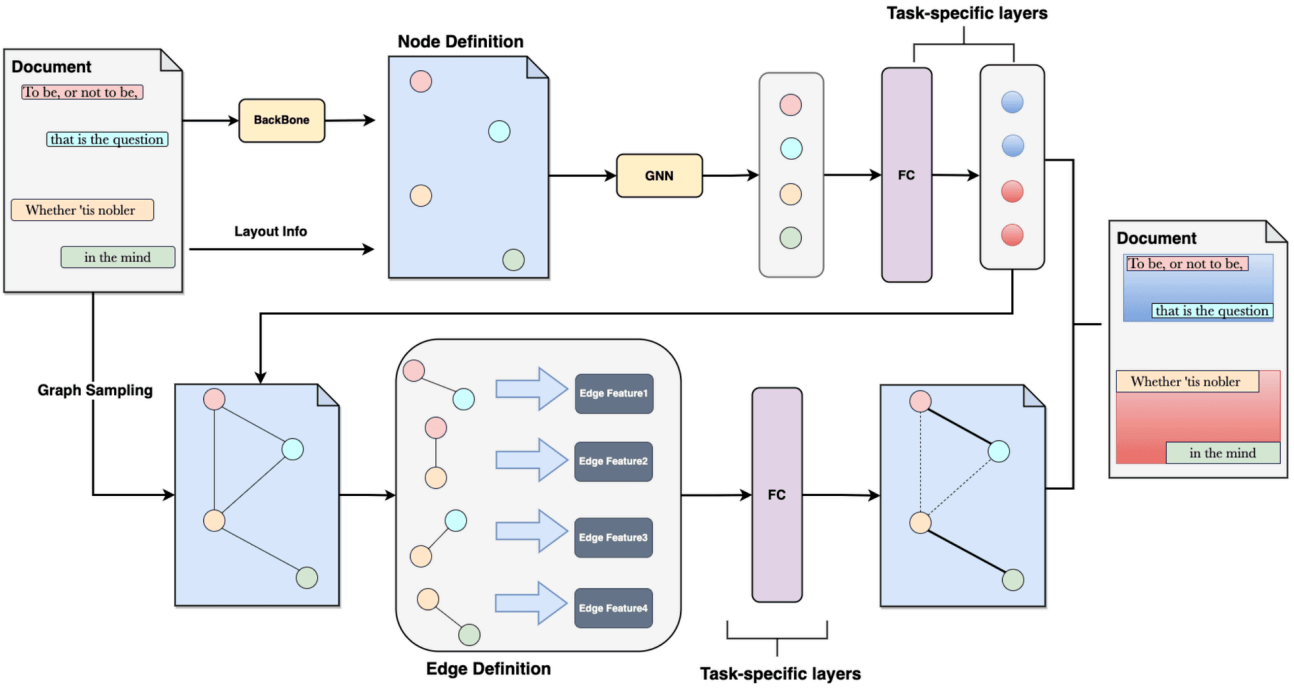


Figure 6: Paragraph2Graph pipeline Training & Evaluation

Dataset

For training and evaluation, three datasets are utilized.

Initially, **DocLayNet** is used for training and testing the models. DocLayNet is one of the benchmark datasets for Document Layout Analysis. It is structured similarly to the benchmark dataset PubLayNet. The PubLayNet dataset consists of pages taken from a collection of medical papers, while DocLayNet has pages collected from different types of documents: financial reports, manuals, scientific articles, laws & regulations, patents, and government tenders. Since PubLayNet uses only medical papers, even though there may be some differences, academic papers still contain some structure. That is why the DocLayNet dataset is chosen for training, as it includes various types of documents with different structures, including some unstructured documents such as manuals.

The DocLayNet dataset consists of 80,863 pages from the mentioned document types, each annotated manually by humans. The DocLayNet dataset is entirely in English. Dataset annotations are in COCO (Common Objects in COntext) format, which is also used for object detection tasks for images.Since our target use case is to gather insights from ESG reports and IFRS documents in Turkish, a smaller dataset has been prepared to test the success of graph-based methods. For this reason, models trained with DocLayNet are also utilized due to the limited number of annotated documents required for the models to converge. However, this approach allows us to see the initial feasibility of graph-based methods and test language agnostic capabilities of graph based methods since they mostly rely on language independent features layout graphs.

For annotating Turkish documents, the LabelBox platform is used. This platform allows easier manual annotation by using keyboard shortcuts and having the ability to label connections with reading order and tag related components of the pages. An annotation of a page can be done within approximately 3 minutes on average, though for more complicated pages with several components, this can be up to 10 minutes or more. A total of 822 pages from ESG and IFRS reports are annotated.

As can be seen in Figure 6, an ESG report page can contain multiple components that are ordered in an unstructured way. There are multiple images in the page which are generally not needed for information extraction but they overlap with text blocks which can be problematic especially for image-based approaches.

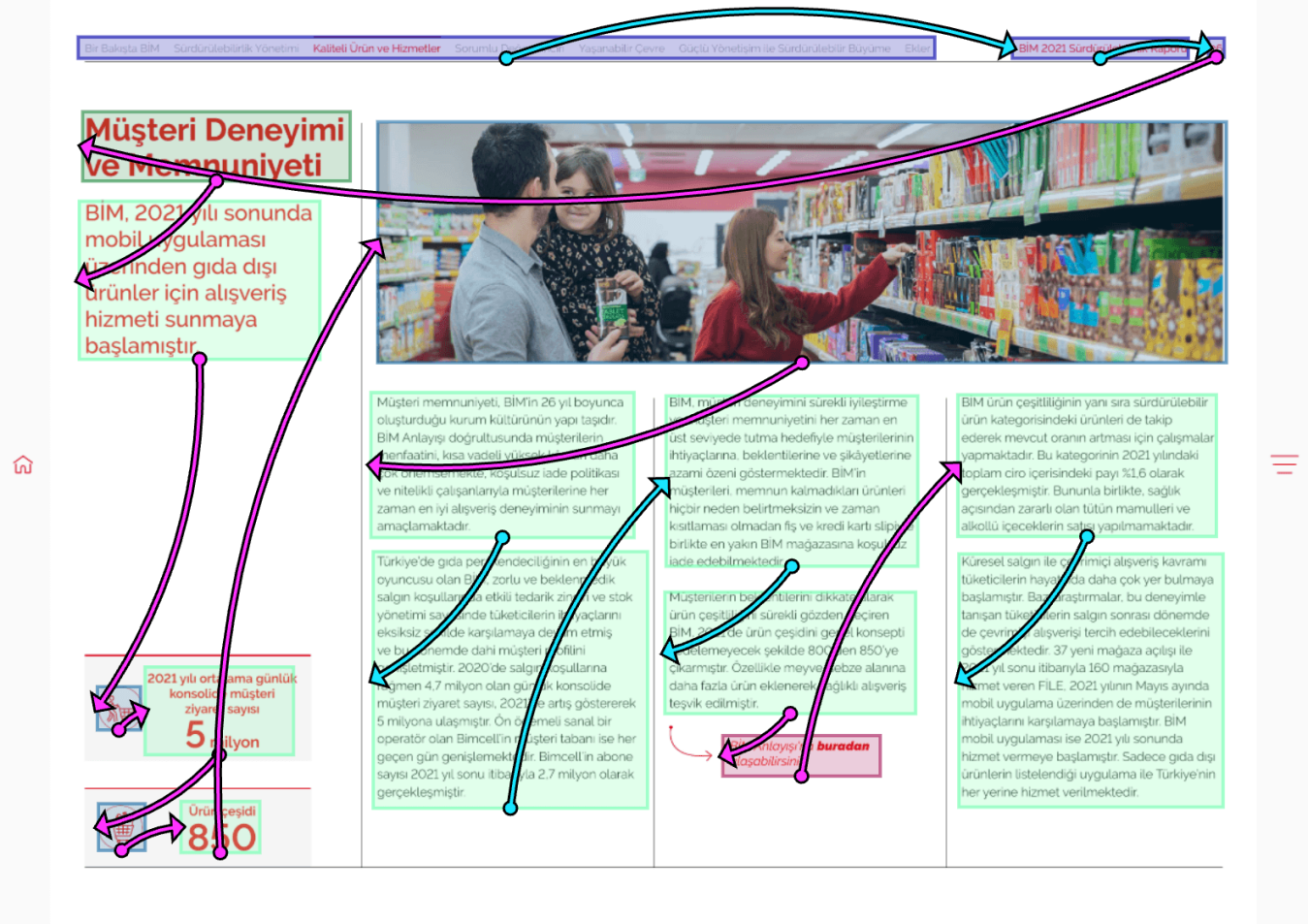
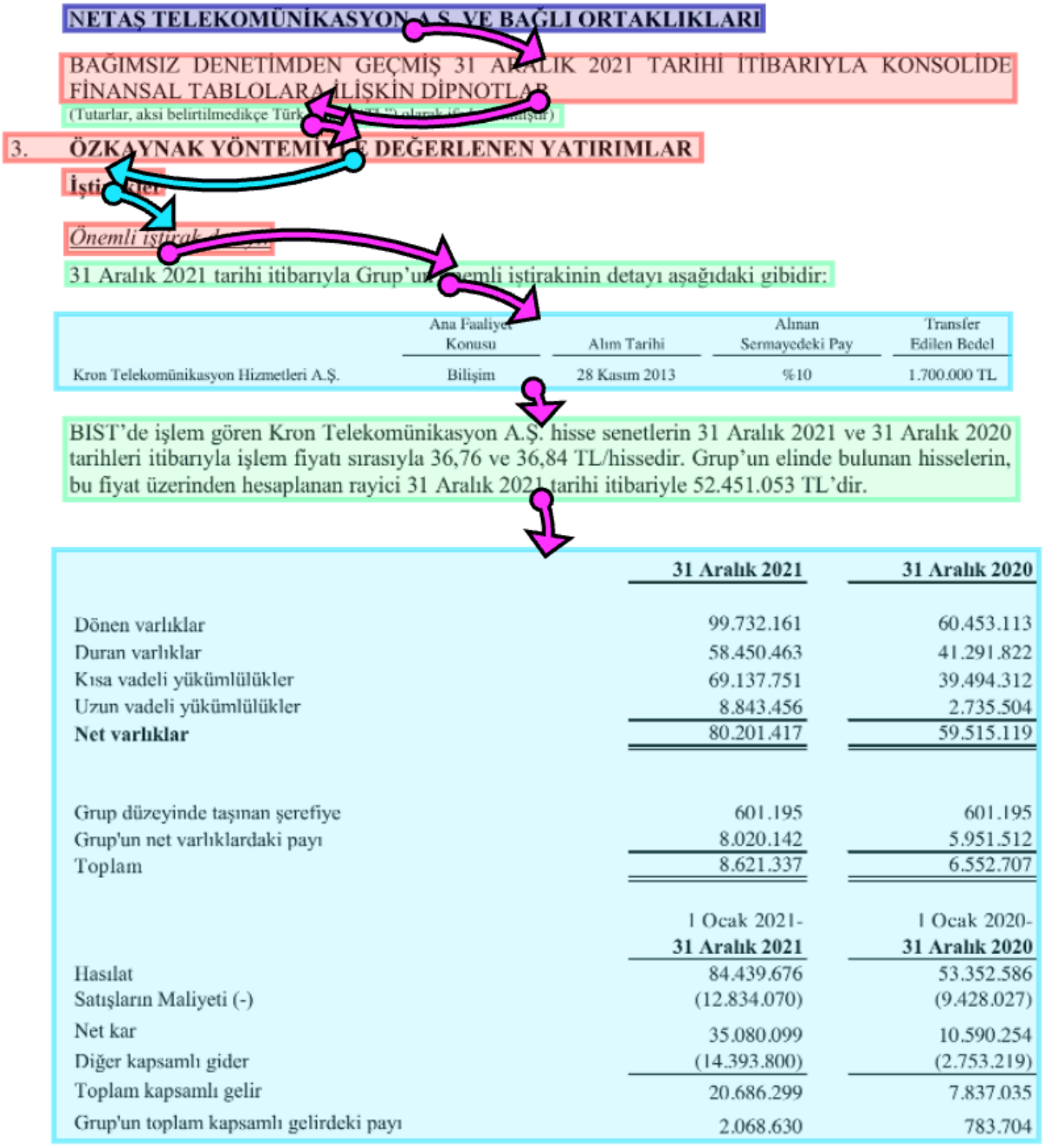


Figure 7: Annotated ESG report page – Blue arrows indicate connected components, purple arrows indicate independent components. The direction of the arrows indicates the reading order.

In Figure 7, an annotated IFRS page is shown. While IFRS adheres to a standard and layout components are easier to detect, it has its own challenges such as the additional piece of text after table header and before the values in the table.



NETAŞ TELEKOMÜNİKASYON A.Ş. VE BAĞLI ORTAKLIKLARI

BAĞIMSIZ DENETİMDEN GEÇMİŞ 31 ARALIK 2021 TARİHİ İTİBARIYLA KONSOLİDE FİNANSAL TABLOLARA İLİŞKİN DİPNOTLAR

(Tutarlar, aksi belirtilmedikçe Türk

3. ÖZKAYNAK YÖNTEMİYLE DEĞERLENEN YATIRIMLAR

İşti

Önemli iş

31 Aralık 2021 tarihi itibarıyla Grup'un emli iştirakinin detayı aşağıdaki gibidir:

	Ana Faaliyet Konusu	Alım Tarihi	Alınan Sermayedeki Pay	Transfer Edilen Bedel
Kron Telekomünikasyon Hizmetleri A.Ş.	Bilişim	28 Kasım 2013	%10	1.700.000 TL

BIST'de işlem gören Kron Telekomünikasyon A.Ş. hisse senetlerin 31 Aralık 2021 ve 31 Aralık 2020 tarihleri itibarıyla işlem fiyatı sırasıyla 36,76 ve 36,84 TL/hissedir. Grup'un elinde bulunan hisselerin, bu fiyat üzerinden hesaplanan rayici 31 Aralık 2021 tarihi itibariyle 52.451.053 TL'dir.

	31 Aralık 2021	31 Aralık 2020
Dönen varlıklar	99.732.161	60.453.113
Duran varlıklar	58.450.463	41.291.822
Kısa vadeli yükümlülükler	69.137.751	39.494.312
Uzun vadeli yükümlülükler	8.843.456	2.735.504
Net varlıklar	80.201.417	59.515.119
Grup düzeyinde taşınan şerefiye	601.195	601.195
Grup'un net varlıklardaki payı	8.020.142	5.951.512
Toplam	8.621.337	6.552.707

	1 Ocak 2021- 31 Aralık 2021	1 Ocak 2020- 31 Aralık 2020
Hasılat	84.439.676	53.352.586
Satışların Maliyeti (-)	(12.834.070)	(9.428.027)
Net kar	35.080.099	10.590.254
Diğer kapsamlı gider	(14.393.800)	(2.753.219)
Toplam kapsamlı gelir	20.686.299	7.837.035
Grup'un toplam kapsamlı gelirdeki payı	2.068.630	783.704

Figure 8: An example annotated page from the IFRS report.

Experiments & Evaluation

The training for models is conducted on DocLayNet using a single core of an RTX 5000 GPU, with a training split. Training for the GLAM network took about 2 hours, while Paragraph2Graph took about 48 hours to converge.

The first part of the evaluation is performed using mean average precision (mAP) scores based on the bounding boxes of detected layout components, as determined by the COCO evaluator. Additionally, an empirical analysis will be presented comparing results from Graph-based methods with those of LayoutLMv2, a text-based DLA model developed by Microsoft, which is considered one of the state-of-the-art models. The LayoutLMv2 model used is fine-tuned specifically for Turkish documents. This comparison is made with the purpose of analyzing how the results differ for these two approaches and what advantages if any can graph-based models provide.

The tables below present results for three different test sets of DocLayNet and Turkish documents.

Method	DocLayNet Testset		Turkish Documents	
	Paragraph2Graph	GLAM	Paragraph2Graph	GLAM
Text	**0.811**	0.764	**0.761**	0.689
Table	0.827	**0.868**	0.794	**0.909**
Title	0.643	**0.841**	0.454	**0.820**
Page-Header	**0.796**	0.771	**0.815**	0.687
Page-Footer	**0.920**	0.850	**0.648**	0.521
Section Header	**0.824**	0.794	**0.875**	0.711
Footnote	0.625	**0.706**	0.564	0.403
Caption	**0.809**	0.743	**0.724**	0.514
Total	0.778	**0.808**	**0.635**	0.614

Table 1: mAP scores for DocLayNet test set and Turkish documents

Paragraph2Graph is shown to capture certain layout elements, such as headers and footers, more effectively. While GLAM also has success with headers and footers, it seems to have struggled with identifying them from each other, especially in complex pages such as those in ESG documents. Even though the training is done with an English dataset, results in Turkish dataset still has shown considerable performance, indicating the language agnostic capabilities of graph neural networks for layout analysis.

Example Results & Discussion

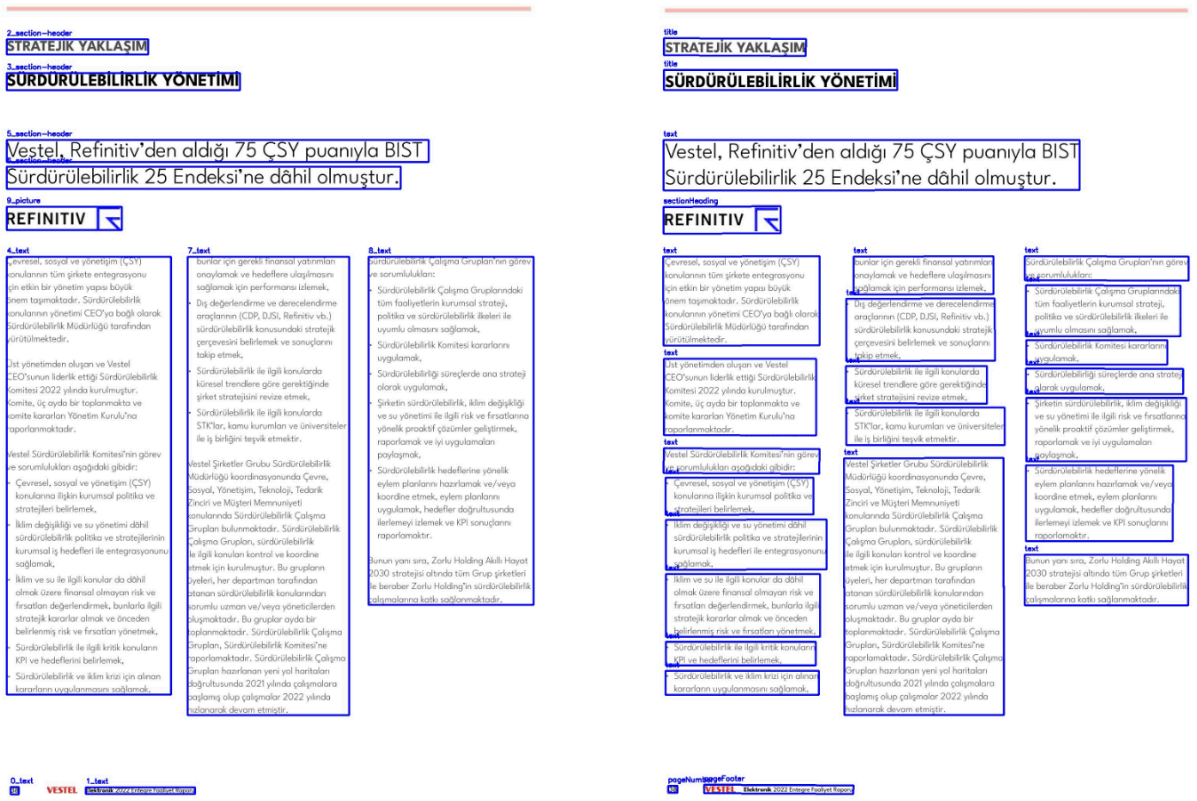


Figure 9: On the left, results from the Graph-based method, and on the right, results from an SOTA model

Figure 9 demonstrates a comparison between graph-based layout models trained in this study and the LayoutLMv2 model for the same page from a Turkish ESG document. Both models are shown to have similar success, correctly identifying the layout elements with mostly correct categories assigned, except for small mistakes. There are two points to note with this example. Initially, the Graph-based model is trained solely on an English dataset, while the other model used is one specifically tailored for Turkish documents. Secondly, while the LayoutLMv2 model successfully identifies each paragraph, in a document like this, where there are three columns of text, additional work is required to sort these paragraphs into a meaningful reading order to use in downstream tasks. Results from the graph-based model indicate that the three columns of text are clearly distinct. Also, graph-based methods work by merging smaller elements into larger layout elements depending on edge weights between them, so it is possible to retrieve smaller paragraphs from text columns in a trivial way.

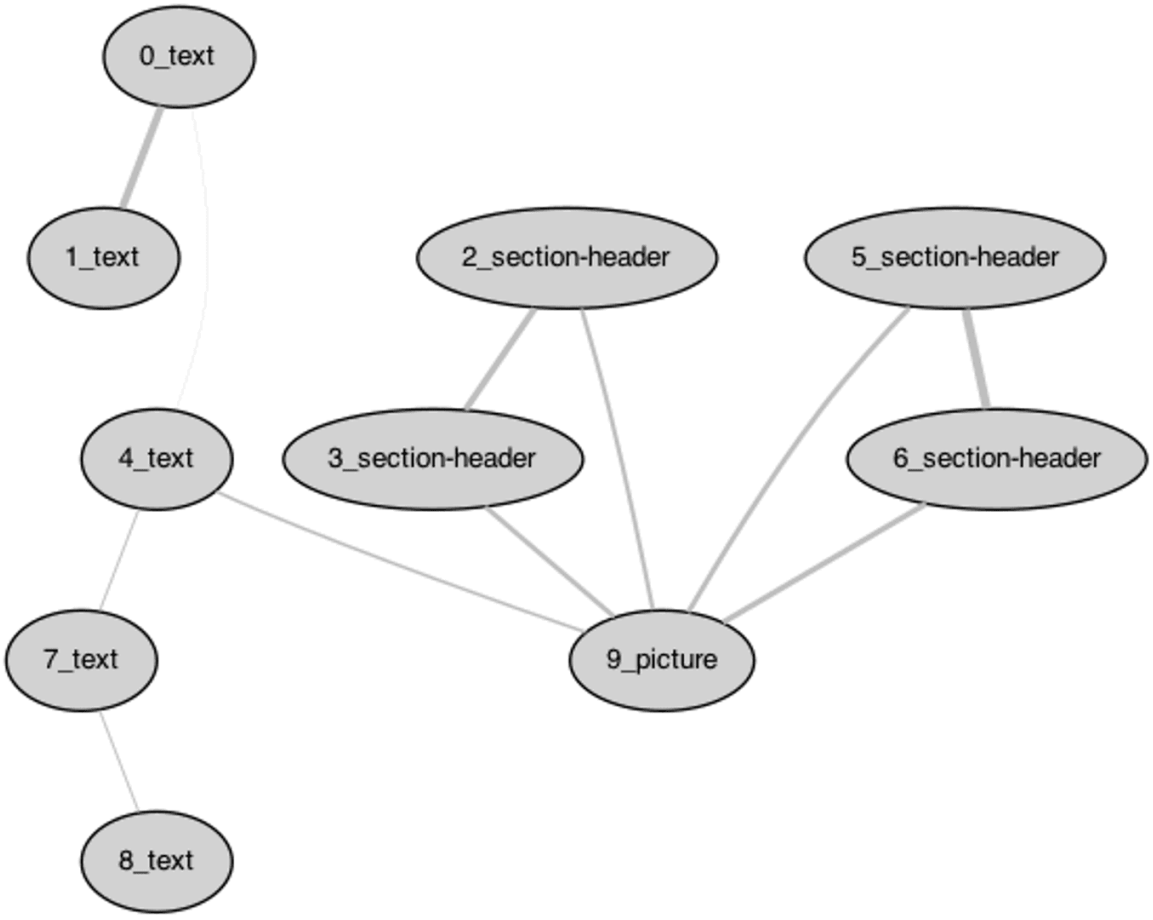


Figure 10: Graph constructed for layout analysis. Only edges with high weights are shown.

Figure 10 shows the graph for the respective layout elements for the image on the left. The connections between three text columns show the reading order is preserved.

Figure 11: Side-by-side comparison of graph-based (left) and SOTA model (right).

Figure 11 shows another comparison between the two models. Again, two models show similar success in extracting layout elements. One noticeable detail is the identification of a rather unusual graphical representation of information on the page. It shows a column graph with values written on columns. The LayoutLMv2 model successfully identifies each value without connecting them. On the other hand, graph model puts the whole graph into a singular element with table tag, potentially easier retrieval of these values.

Conclusion

In this study, graph neural networks have been explored as a method for Document Layout Analysis, serving as an alternative to state-of-the-art vision-based and language model-based methods in this field. While its performance metrics are within range of those models, they do not outperform those. However, results on unstructured documents show that there are some advantages to using graph-based methods in some use cases. First of all, graph-based methods are language agnostic as they are shown to work well for Turkish documents despite being solely trained on English-language documents. Additionally, when it is required combine various layout elements for downstream tasks, or when reading order is required this information can be inferred from nodes and edges in graph structure while for other models such a task may require an additional layer of classifier or post-processing step.

Given the banking sector's reliance on complex, document-driven workflows, utilizing large language models (LLMs) becomes an essential strategy. The use of retrieval-augmented generation (RAG), which directly leverages these documents, is the most reliable approach to achieve this. In this respect, layout detection becomes essential for partitioning documents into smaller, meaningful parts for easy and efficient retrieval in RAG systems. In this respect, graph-based models are shown to offer advantages in the easier identification of connected layout elements and coherent reading order among these. Also, their language-agnostic capabilities allow them to be deployed in regions with a low availability of documents.

It can be concluded that graph neural networks can be a viable alternative to image-based and text-based methods for document layout analysis, especially while working with limited-resource languages and unstructured documents. Also, graphs offer advantages in identifying connections between layout elements, thus making them a viable candidate for RAG systems.

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